ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · W1 · WORKING SCIENTIFICALLY

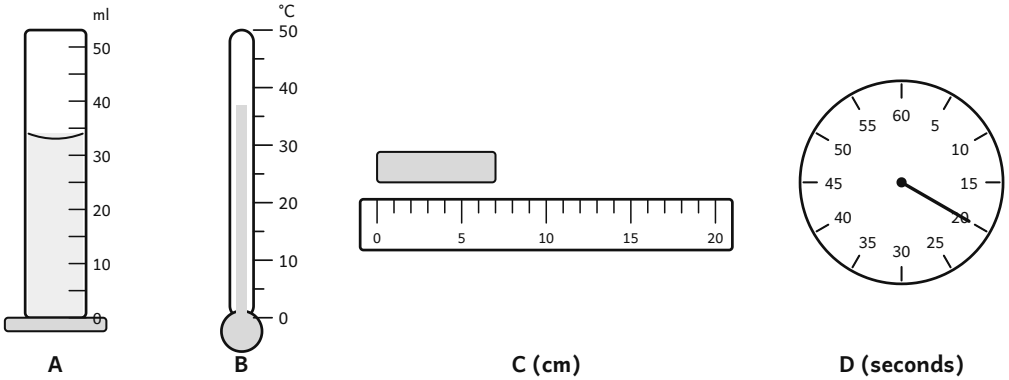
Tools, units and tests

★☆☆
Grade 3

Learning goals

- Match tools and units.
- Spot testable questions.

1 Read each instrument. Include the unit. [4]



A B C (cm) D (seconds)

A	B	C	D
34 ml	37 °C	7 cm	20 s

2 Draw a line to match the tool to the unit. [4]

Draw a line to match each pair, or write the letter.

thermometer ●	→ A	● **A** °C
stopwatch ●	→ C	● **B** cm
ruler ●	→ B	● **C** seconds
forcemeter ●	→ D	● **D** newtons

3 Can it be tested? Sort the questions. [4]

SORT THESE

Which ball bounces highest? **Does salt melt ice faster?**
Is red better than blue? **Which song is the nicest?**

Can be tested	Cannot be tested
Which ball bounces highest?	Which song is the nicest?
Does salt melt ice faster?	Is red better than blue?

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · W1 · WORKING SCIENTIFICALLY

Making it fair

★★☆
Grade 3

Learning goals

- Plan a fair test.
- Explain repeats.

1 **Lina wants to know which cup keeps cha hot longest. She puts hot cha in a clay cup and cold cha in a steel cup. Is it fair? How can she fix it?** [3]

No: the starting temperature is different. She must start both with the same amount of cha at the same temperature.

2 **Why should you repeat a measurement three times?** [2]

To check the results and make them more reliable.

3 **Write a prediction with a reason: "Will a plant grow taller in sunlight or in shade?"** [2]

I think it will grow taller in sunlight because plants need light to make food.

ANSWER KEY

INVESTIGATE · W1 · WORKING SCIENTIFICALLY

Which paper makes the best paper plane?

★★☆ Grade 3

Learning goals

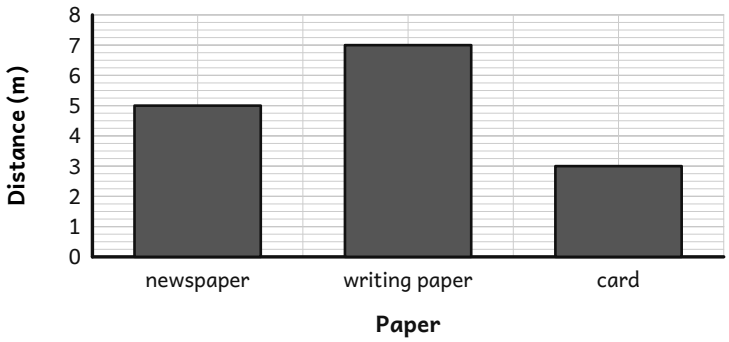
- Repeat measurements.
- Draw a bar chart.
- Say what the evidence shows.

Class 3A made the same paper plane from three kinds of paper and threw each one three times.

1 **Complete the table. Find the middle (median) result of each.** [3]

Paper	Try 1 (m)	Try 2 (m)	Try 3 (m)	Middle result (m)
newspaper	4	5	6	5
writing paper	6	8	7	7
card	3	2	4	3

2 **Draw a bar chart of the middle results.** [3]



3 **Which paper was best? What does the evidence show?** [2]

Writing paper flew furthest (7 m), so it makes the best plane.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · W1 · WORKING SCIENTIFICALLY

Testing umbrellas for the monsoon

★★★ Grade 3

Learning goals

- Plan a real test.
- Use evidence to choose.

Science | Design | Maths

The monsoon rains are coming. A shopkeeper in Rajshahi wants to sell the best umbrellas. He asks a Grade 3 class to test three kinds of cloth: cotton, plastic sheet and nylon.

The class drips 50 drops of water onto each cloth stretched over a cup, then measures how much water gets through.

A story about a class investigation. Results are example data.

1 **Complete the plan.** [3]

Change	the type of cloth
Measure	water that gets through (ml)
Keep the same	number of drops, size of cloth, cup

2 **Results: cotton 8 ml, plastic 0 ml, nylon 1 ml. Which cloth should the shopkeeper use? Why?** [2]

Plastic sheet (or nylon): almost no water gets through.

3 **What else should they test before choosing?** [1]

Strength, weight, cost, whether it tears in wind.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · W1 · WORKING SCIENTIFICALLY

Show what you know

★★★
Grade 3

Learning goals

- Use science words.

1 Find the words.

K	E	K	R	E	P	E	A	T
F	T	E	B	Y	J	A	I	C
A	H	K	I	W	F	B	T	H
I	G	P	T	E	S	T	A	A
R	D	P	B	R	O	R	B	R
P	R	E	D	I	C	T	L	T
F	J	P	G	G	D	Y	E	T
H	M	E	A	S	U	R	E	J
U	T	H	M	W	A	L	W	S

chart fair measure predict repeat table test

2 Which tool measures time? [1]

○ thermometer
● **stopwatch**
○ ruler

3 In a fair test you change... [1]

● **one thing**
○ two things
○ everything

4 Challenge. Plan a test: "Which ball rolls furthest?" Write what you change, measure and keep the same. [3]

Change	the type of ball
Measure	distance rolled (cm)
Keep the same	ramp, push (none), floor